



BIOLOGY CH.5 — REPRODUCTION IN ANIMALS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway



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1. Puberty & Adolescence — Changes & Definitions

- Puberty (12-16 boys, 10-14 girls) mein sexual maturation se juda change 'mammary glands ki growth' hai (girls mein) — hair growth, appearance consciousness, pimples generic changes hain. [Q173]
- ⚠️ Trap: Sex education dene se teenagers ek-doesre ka mazaak NAHI banate — ye galat reasoning hai. Sex education STDs se awareness, sexual abuse se bachaav, aur myths dur karti hai. [Q176]
- ⚠️ Trap: 'Obesity' pubertal development ka part NAHI hai — pimples, menstruation, voice change puberty ke normal changes hain. [Q184]
- Adolescence 11 saal ki age se shuru hokar 18-19 saal tak chalti hai — rapid physical-cognitive-psychosocial growth ka time hota hai. [Q216]
- Adolescent ko healthy reproductive system develop karne ke liye 'nutritionally well-balanced' diet chahiye — WHO ke mutabik reproductive health mein behavioural-emotional-physical-social wellbeing shaamil hai. [Q232]
- Adolescence wo age hai jisme reproductive organs functional ho jaate hain, gonads gametes banana shuru karti hain. [Q283]
- Jab reproductive tissues mature hoti hain aur body ki normal growth slow ho jaati hai, is state ko 'Youth' (Puberty) kehte hain. [Q287]
- Boys mein puberty ke time appearance changes (beard growth, hoarse voice, reproductive organs development) 'Testosterone' hormone se aate hain. [Q301]
- Females mein puberty ki shuruaat 'Menarche' (pehla menstrual period) se hoti hai. [Q321]

2. Sex Determination — Environmental vs Genetic

- 'Snail' mein sex genetically determined NAHI hoti — Sparrow, Butterfly, Moth mein genes-chromosomes se sex determine hoti hai. [Q175]
- Kuch reptiles mein fertilized eggs ka 'Temperature' unke gender determine karta hai — Temperature Dependent Sex Determination (TSD). [Q183]
- Kuch animals mein gender determination ka environmental factor 'Temperature' hai — lizard jaisa example, jabki humans mein ye genetically hota hai. [Q185]
- 'Reptiles' class mein temperature kuch species ka sex determine karti hai — crocodilians, kai turtles-lizards mein TSD dekhi jaati hai. [Q206]
- Snail apni life mein sex change kar sakta hai (male se female) — isse dikhta hai ki sex genetically NAHI, environmental-stress conditions se bhi determine ho sakti hai. [Q214]
- Reptiles mein offspring ka sex 'fertilized egg jis temperature par rakha jaata hai' usse determine hota hai — 30°C se zyada temperature par females, 25°C se kam par males bante hain. [Q219]
- 'Protogyny' wo process hai jisme organism female ki tarah life shuru karke baad mein male mein transform hota hai (jaise Banana, Fig, Plum flowers mein) — Protandry ismein ulta hota hai. [Q208]

3. Asexual Reproduction — Types & Examples

- ⚠️ Trap: 'Pollination' asexual reproduction ka example NAHI hai — ye flowering plants ki sexual reproduction process hai. Grafting, Binary Fission, Budding asexual reproduction ke examples hain. [Q174]
- Amoeba mein do individuals mein divide hone ki process 'Binary Fission' kehlaati hai — Amoeba isi tarah reproduce karta hai. [Q239, Q259]
- Hydra budding aur regeneration dono kar sakta hai — Phylum Coelenterata se belong karta hai. Budding mein bud se naya organism develop hota hai (Bacteria, Yeast, Corals mein bhi hoti hai). [Q253]
- Hydra, starfish, planaria 'Regeneration' process se guzarte hain — kho gaye ya amputate hue body parts replace/restore karte hain. Fragmentation mein organism pieces mein toot jaata hai (Fungi, lichens). [Q272]
- Fully matured tiny individuals jab parent body se detach hokar naye individual bante hain, ye 'Budding' hai — example: Jellyfish. Binary fission (Amoeba) mein cell 2 parts mein split hoti hai. [Q273]
- Asexual reproduction 'Lower animals aur plants' mein hoti hai — Protozoans, Sponges, Planaria examples hain. [Q266]
- Plasmodium 'Multiple fission' dikhata hai — ek parent cell simultaneously kai daughter cells mein divide hota hai. [Q282, Q297]
- Asexual reproduction mein naye organisms parent ke 'genetically identical' hote hain — sexual reproduction se ye alag hai (jisme gametes ke nuclei fuse hote hain). [Q306]

4. Sex Determination in Humans — Chromosomes (XX/XY)

- Autosomes 22 pairs kehlaate hain ($2n=18$ wale organism mein 16 autosomes) — sex-determination se unrelated genes carry karte hain. Humans mein 22 autosome pairs aur 1 sex chromosome pair hota hai. [Q178]
- Baby ka sex sperm mein present chromosome (X ya Y) se determine hota hai — har egg mein X chromosome hota hai, sperm X ya Y ho sakta hai. Isliye male hi human sex determination ke liye responsible hota hai. [Q189, Q295, Q270]
- Sex determination flow chart se conclude hota hai ki 'The female decides the sex of the offspring' galat hai — actually 'Male decides' (Y chromosome se boy, X chromosome se girl) — sabhi children apni maa se X chromosome inherit karte hain. [Q191, Q193]
- Male gametes (sperm) mein half mein X chromosome hota hai ($22+X$), half mein Y chromosome ($22+XY$ nahi, sirf $22+Y$) — female gametes sirf X chromosome carry karte hain. [Q194]
- Allosomes (sex chromosomes) sex determination ke liye responsible hote hain — Autosomes (22 pairs) sex determination se unrelated hote hain. [Q209]
- Statement A ('sabhi children apni maa se X chromosome inherit karte hain'), C aur D sahi hain — statement B galat hai kyunki X inherit karna dono (boys-girls) ke liye hota hai. [Q212]
- ⚠ Trap: 'Sabhi children apni maa se Y chromosome inherit karte hain' — ye statement INCORRECT hai — maa sirf X chromosome pass karti hai, Y chromosome sirf pita se milta hai. [Q217]
- ⚠ Trap: X-containing sperm aur Y-containing sperm mein size ya motility ka koi visible difference NAHI hota. [Q223]
- Human women mein 'perfect pair' (XX) sex chromosomes milte hain — men mein mismatched pair (XY, ek badi X aur ek chhoti Y) hota hai. [Q229]
- X-chromosome wale sperms aur Y-chromosome wale sperms — dono mein 23 chromosomes hote hain (equal count, sirf sex chromosome type alag). [Q230]
- ⚠ Trap: 'Y-chromosome' sirf human males mein paya jaata hai — females sirf X chromosomes rakhti hain (XX). [Q264]
- Human male 'X sperm' apni maa se receive karta hai — pita se X ya Y sperm banane ki capacity milti hai, maa se sirf X chromosome milta hai (jo har egg mein hota hai). [Q289, Q271, Q237]
- Female mein XX set of sex chromosomes hote hain — male mein XY. [Q275, Q240]
- Girl child ke chromosomes '44 autosomes + XX' hote hain — boy child ke '44 autosomes + XY'. [Q318]
- Diploid organisms (separate sexes) mein ek specific pair of chromosomes sex determine karta hai — inhe 'Sex Chromosomes' kehte hain. [Q322]

5. Chromosome Numbers — General Counts

- Human germ cell (sperm ya egg) mein 23 chromosomes hote hain (46 ka half) — normal human cell mein 46 chromosomes. [Q220, Q277, Q284]
- Normal human mein 22 pairs autosomes hote hain (44 total) — 23rd pair sex chromosome hoti hai. [Q235]
- 'Homologous chromosomes' autosomes hote hain — same gene sequence-loci-length rakhte hain, males-females mein same dikhte hain. [Q242]

- Human somatic cells mein 46 chromosomes hote hain (23 pairs) — sperm-egg ke alawa sab body cells mein. [Q268]
- Rat mein 21 pairs (total 42) chromosomes hote hain — doosre animals: Human (46), Pea (14), Cat (38), Dog (78). [Q249]

6. Genes, DNA & Heredity — Basic Features

- Reproduction ka basic event 'DNA replication' hai — daughter cells ko DNA ki identical copy milti hai, parents se offspring tak information transfer karti hai. [Q186, Q292]
- ⚠ Trap: Genes 'koi change nahi hote' — ye galat statement hai (genes segments of DNA hain, jo change ho sakte hain — mutations). Genes proteins produce karke traits control karte hain, DNA ke segments hain. [Q179]
- ⚠ Trap: Human female mein fertilization 'uterus mein' NAHI hoti — ye galat statement hai, actually fallopian tubes mein hoti hai. [Q187]
- Progeny mein minor differences 'DNA copying mein errors' se aati hain — asexual reproduction mein bhi small inaccuracies se variations aati hain. [Q188]
- Non-dividing cell mein DNA 'Chromatin content' ki tarah exist karta hai — divide hone se pehle chromatin condense hokar chromosome banata hai. [Q192]
- Trait inheritance 'Paternal aur Maternal DNA' se govern hota hai — dono parents se DNA milti hai. [Q201]
- Genes traits ko 'Hormones' ki activity ke through control karte hain — genes proteins-hormones produce karte hain jo traits determine karte hain. [Q205]
- Sexually reproducing individuals ke paas ek hi trait ke 2 copies (genes) hoti hain — ek dominant, ek recessive express ho sakti hai. [Q238]
- Genes ko DNA sequence mein mark karne ki process 'Annotation' kehlaati hai — genomic data ko correct genes-locations tak map karna. [Q245]
- Reproduction 2 broad categories mein divide hoti hai — Sexual aur Asexual reproduction. [Q260]
- Sexual reproduction mein parent apne 'half genes' contribute karta hai — dono parents equally genes dete hain zygote ko. [Q265, Q309]

7. Sexual Reproduction & Genetic Diversity

- Sexual reproduction sabse zyada diversity la sakti hai — gametes ka fusion 2 alag individuals ke genes combine karta hai, genetic diversity badhati hai. Asexual reproduction mein offspring genetically identical hote hain. [Q256, Q263, Q280, Q314]

8. Cell Division — Mitosis & Meiosis

- Diagram mein A-Mitosis (body cells banane ke liye) aur B-Meiosis (gametes banane ke liye reproductive cells mein) hota hai. [Q180]

- Chromosome number generation-to-generation maintain hone ke liye 'gamete formation ke dauran chromosomes halve hote hain' (Meiosis) — reduction division daughter cells ko half chromosomes deti hai. [Q198]
- ⚠ Trap: Meiosis se chromosomes 'double' NAHI hote — ye galat statement hai, actually halve hote hain. Meiosis 2 phases mein hoti hai, gonads mein gametes production ke liye. [Q317]
- 'Cytokinin' cell division se concerned hai — cell division growth, tissue repair, gametes production ke liye crucial hai. (Auxin growth control, Gibberellin dormancy break karte hain.) [Q319]

9. IVF (In Vitro Fertilization) & Test-Tube Babies

- Blocked oviducts wali women ke liye 'IVF' (zygote outside kept, test tube mein development) solution hai — fertilization glass vessel mein egg-sperm combine karke hoti hai. [Q181]
- IVF mein 'external fertilization of sperm and ovum' hoti hai — pre-embryos wapas uterus mein transfer hote hain. Patrick Steptoe aur Robert Edwards ne IVF devise ki. [Q195]
- 'Test tube baby' mein fertilization test tube mein hoti hai, lekin development uterus mein hota hai — Louise Joy Brown duniya ki pehli test-tube baby thi. [Q246]

10. The Male Reproductive System

- Human males mein testes abdominal cavity ke bahar 'scrotum' mein hote hain — body temperature se 2-3 degree kam temperature maintain karta hai sperm formation ke liye. [Q182, Q316]
- Sperms ka path: Testis → Vas deferens → Urethra → Penis — Epididymis testicles se sperm move karta hai. [Q190]
- ⚠ Trap: 'Fallopian tubes block karna' male sterilization method NAHI hai — 'Vas deferens block karna' (vasectomy) male sterilization ka surgical method hai. [Q200]
- Sperm ka Head (Part A) zygote ka sex determine karta hai — nucleus with chromosomes contain karta hai. Middle piece mitochondria energy deti hai, Tail flagellum motility deta hai. [Q228]
- Male reproductive cells 'testes' mein produce hote hain — spermatogenic cells seminiferous tubules mein hote hain. [Q252, Q298, Q308]
- ⚠ Trap: 'Ovary' male reproductive system ka part NAHI hai — Vas deferens, Testis, Scrotum male parts hain. [Q258, Q304]
- Sperm aur urine dono ke liye common passage 'Urethra' hai — urinary bladder se penis tak extend hoti hai. [Q281]
- Human males ke 2 testicles hote hain — sperm aur testosterone produce karte hain. [Q285]
- Male reproductive system ka main genetic material 'Sperm' mein milta hai. [Q290]
- Testicle se sperm upward le jaane wala duct 'Sperm duct' (Vas deferens) kehlaata hai. [Q291, Q296]
- Sperms temporarily 'Epididymis' mein store hote hain — maturation ke liye. [Q302]

11. Types of Germ Cells — Gametes

- Complex body wale sexually reproducing animals mein — Male germ cell 'small aur motile' hota hai (sperm), Female germ cell 'large with food store' hota hai (ovum). [Q199]
- Reproductive cell 'Gamete' kehlaata hai — Lymphocytes, RBC, WBC reproductive cells NAHI hain. [Q241]
- Food store rakhne wala germ cell 'Female gamete' (ovum) kehlaata hai — non-motile hota hai. Male gamete (sperm) motile hota hai. [Q257]
- Motile germ cell 'Male gamete' (sperm) kehlaata hai — female egg tak fertilize karne jaata hai. [Q313]

12. Fertilization & Zygote Formation

- Single-celled fertilised egg 'Zygote' kehlaata hai — sperm-egg combine hone se banta hai. (Blastocyst dividing cells cluster hai, Embryo initial development stage hai.) [Q236]
- Embryo lagataar grow karke organs develop karta hai aur 'foetus' banta hai — Placenta oxygen-nutrients deti hai, Zygote fertilized egg cell hai. [Q221]
- Male aur female sex cells fuse hokar 'Zygote' banate hain. [Q248]
- 2 alag bacterial cells sexual reproduction ke dauran combine hokar 'Zygote' banate hain. [Q254]
- 'Fertilization' se Zygote ki formation hoti hai — male aur female gamete ka fusion. [Q261, Q276]
- 'Syngamy' sperm aur egg ke fusion ko kehte hain — angiosperms mein double fertilization ka essential feature hai. [Q310]

13. Contraception Methods

- Oral contraceptive pills 'ovulation aur fertilization' dono rokhti hain. [Q177]
- Permanent sterilization method 'Surgical methods' hain (tubal sterilization females mein, vasectomy males mein) — condom, birth control pills, copper-T non-permanent hain. [Q196]
- ⚠ Trap: Contraceptive devices 'no side effects' rakhte — ye statement INCORRECT hai. Ye pregnancy prevent karte hain, population control karte hain, lekin side effects ho sakte hain. [Q204]
- Condom pregnancy aur STDs (HIV samet) dono prevent karta hai — surgical method, copper-T, oral pills sirf pregnancy prevent karte hain. [Q211]
- Loop (IUD) contraceptive ki tarah 'fallopian tubes ka access block' karta hai — copper-coated IUD sperm-egg fertilization rokta hai. [Q227]
- Contraceptive pills ka disadvantage hai — 'hormonal imbalance' produce kar sakte hain. [Q231]
- 'Pills ka intake' hormonal balance change karta hai — Copper IUD copper release karta hai, Hormonal IUD progestin release karta hai. [Q233]
- Copper-T uterus mein place karne ki method 'Contraception' kehlaati hai. [Q293]

14. Reproductive Health — Myths & Facts

- ⚠️ Trap: 'Chips-canned snacks avoid karna chahiye adolescence mein' — ye MYTH NAHI, FACT hai. 'Menstruating girl ko boys nahi dekhne chahiye' aur 'kitchen mein kaam nahi karna chahiye' — ye myths hain. [Q197]
- ⚠️ Trap: 'Unborn child ka sex determine karna' Reproductive Health ka feature NAHI hai — STD prevention, family planning, reproductive process knowledge iske features hain. [Q210]
- ⚠️ Trap: Menstrual flow ke time 'chips-packed snacks khaana' nutritional needs पूरी nahi karta — ye FALSE statement hai. Sanitary napkins use karna, cycle track karna, 4-5 ghante mein pads badalna sahi practices hain. [Q222]

15. Mendel & Heredity

- Gregor Mendel ne discover kiya ki sexually reproducing organisms mein 'ek pair of factors' unki characteristics define karti hai — pea plants par experiments se dominance-segregation-independent assortment ke laws diye. [Q203]

16. The Female Reproductive System & Menstrual Cycle

- Diagram mein A (Uterus) — implantation-embryo development, B (Fallopian Tube) — egg ki fertilisation. [Q207]
- 'Oestrogen' ovarian hormone hai — female ovaries se secrete hoti hai (Progesterone bhi). Testosterone male hormone hai. [Q213]
- Menstruation ke dauran 'uterine lining shed' hoti hai — pregnancy na hone par uterus apni lining nikalta hai. Hormones control karte hain: LH, FSH, estrogen, progesterone. [Q215]
- Menstruation ki cessation (khatam hona) 'Menopause' kehlaati hai — 45-50 saal ki age mein hoti hai. Menarche pehla period hota hai. [Q202]
- Uterine wall fertilized egg receive karne ki taiyari karti hai — fertilization na hone par thickened lining 'Menstruation' (period) mein shed hoti hai. [Q226]
- 'Graafian follicles' mammal ki ovary mein milte hain — mature hokar ovum release karte hain. [Q247]
- Fertilised egg (zygote) 'Uterus' mein implant hoti hai — Cervix-Vagina alag structures hain. [Q250, Q255, Q299, Q311]
- Vagina 'Cervix' tak jaati hai — cervix uterus ko vagina se connect karta hai. [Q274]
- Female germ-cells (eggs) 'Ovaries' mein bante hain — steroid hormones (estrogen-progesterone) bhi produce karti hain. [Q286]
- Humans mein fertilization 'Fallopian tube' mein hoti hai — ovaries ko uterus se connect karti hai. [Q288]
- Egg ovary se uterus tak 'Fallopian tube' (oviduct) le jaati hai. [Q294, Q312]
- ⚠️ Trap: 'Vas deferens' female reproductive system ka part NAHI hai — Uterus, Ovary, Fallopian tube female parts hain. [Q303]
- Endometrium (uterus ki lining) fertilized egg ko nourish karti hai — embryo 'Uterus' mein develop hota hai. [Q315]

- Ovum production ki process 'Oogenesis' kehlaati hai — ovaries mein hoti hai, meiosis I-II ke through primary oocyte se ovum banta hai. [Q320]

17. Animal Life Cycles — Oviparous, Viviparous & Metamorphosis

- Butterfly life cycle ka teesra stage (larva ke baad) 'Pupa' hai — Egg → Larva → Pupa → Butterfly. [Q234]
- Ande dene wale animals 'Oviparous' kehlaate hain — birds, reptiles, amphibians, fish, arthropods is category mein. [Q278]
- Bachche ko janam dene wale animals 'Viviparous' kehlaate hain — humans, cows, dogs viviparous hain. [Q279]
- Kangaroo underdeveloped children ko janam deta hai — Marsupials group (Wallabies, Koalas bhi) mein pouch mein baby develop hoti hai. [Q300]

18. Pregnancy, Childbirth & the Placenta

- Woman ko birth dene ke liye 'well-equipped hospital mein' delivery best option hai — pregnancy 40 weeks (3 trimesters) ki hoti hai. [Q224]
- 'Maternal Health' mein prenatal, natal aur postnatal period ki care shaamil hai — Maternal Mortality Ratio (MMR) ise measure karta hai. [Q225]
- Placenta embryo se 'waste matter remove karti hai aur nutrition provide karti hai' — dono kaam ek saath karti hai. [Q251]
- Fetus mother ke blood se nutrition 'umbilical cord' ke through leta hai — 2 arteries aur 1 vein Wharton's jelly se surrounded hote hain. [Q262]
- Labour pain 'uterus' ki walls mein contractions se hota hai — largest smooth muscle uterus mein hoti hai. [Q267]
- Baby rhythmic muscle contractions se 'uterus' mein janam leta hai — cervix dilate hokar baby ko birth canal se pass karti hai. [Q269]

19. Other Facts — Mule Hybrid, Twins, Plant Reproduction & Breeding Seasons

- 'Mule' ek hybrid hai jo 'male donkey aur female horse' ke crossing se banta hai — female donkey+male horse se 'Hinny' banta hai. [Q243]
- Fraternal twins 'fertilisation stage' par possible hote hain — 2 alag eggs 2 alag sperms se fertilize hoti hain (dizygotic). Identical twins ek hi zygote ke split hone se bante hain. [Q244]
- Beej se seedling develop karne wala part 'Embryo' hai — is process ko Germination kehte hain. Endosperm embryo ko nourish karta hai. [Q218]
- Growing embryo ke liye nutritive tissue 'Endosperm' hai — angiosperms ke seeds mein embryo ko surround karta hai, starch-protein se nourish karta hai. [Q307]

- Monochronic animals saal mein sirf 'ek baar reproduction' karte hain — specific breeding season hoti hai (mammals, birds, kuch reptiles mein). [Q305]